
Ethical form : Approved

Pat.Ryser-Welch <pat.ryser-welch@open.ac.uk>
To: Josh Oxley <joshoxley.11@gmail.com>

17 May 2026 at 17:19

Dear Joshua Oxley (Student)

Thank you for submitting your project ethics checklist.

I have reviewed the ethical approach to your project with the following outcome: Approved

My comments are below and have been attached to your submission record:

Great improvement.

I would suggest you also consider the LLM generated content using possibly content without the authors autorisation. You should comment on it.

Also, look for email generated by LLM that could identify a person.

Good work :-)

Details of your submission are included below. Please take a screen shot of this email or save as a PDF to upload as Appendix 1 to your TMA01.

We expect your project scope to develop and evolve as you work through the module. If the response to any of the questions you have answered in this checklist changes, you must submit a new ethics checklist.

If you have any queries, please get in touch with me.

Your Project Ethics Checklist:

Submitted: 05/05/2026 11:21

Submission ID: 2830

Please select the modules that your project proposal relates to (select all that apply):
["TM354 - Software engineering", "TM358 - Machine learning and artificial intelligence"]

Proposed Project Title (Draft):

Investigating hybrid machine learning techniques for detection of suspicious phishing content in emails

Describe your project in brief:

What Problem Needs Solving?

Widespread accessibility to LLMs has resulted in phishing attempts to become increasingly sophisticated, containing more compelling language and design. According to Microsoft (2025, pg. 37) "AI-automated phishing emails achieved 54% click-through rates compared to 12% for standard attempts — a 4.5x

increase.” Combined with over 90% of reported cyber attacks originating from phishing (Duarte, 2025), it is clear cybersecurity efforts are failing to adequately combat this evolving threat.

The aim of this project is to employ Natural Language Processing (NLP) AI models combined with industry standard anomaly detection methods. Project outcomes include investigating the effectiveness of implementing a hybrid machine learning approach compared to standalone methods — combining semantic and linguistic analysis with anomaly detection models using various architectures — within the application of identifying and flagging suspicious content indicative of phishing attacks in emails. Beyond statistical scores, best practice detection outlined in the NIST Phish Scale (Dawkins and Jacobs, 2023) requires a multifaceted approach, identifying contextual, linguistic, visual and technical cues, success can be assessed on the models ability to enact the steps within the Phish Scale.

Why is it a Problem?

Email has become an integral part of human communication with an approximated 361.6 billion emails sent per day in 2024 and predicted to further rise by 17% in the next 4 years (Statista, 2023) .

Cybersecurity research estimates 3.4 billion of these messages are phishing (Zensec, 2026) establishing this malicious tactic as an everyday threat. The World Economic Forum estimates 95% of cybersecurity breaches are tied to human error (World Economic Forum, 2022). Perpetrators use phishing with the aim to steal personal data directly, typically by impersonating a trusted person/service, or to infiltrate into secure systems, often with the intent to access secure data.

A single phishing attack can be devastating, news headlines in recent times show current cybersecurity efforts within large corporations are insufficient, with government IDs of service users being breached (Discord, 2025). Expectations for data governance in the General Data Protection Regulations 2016 (GDPR) impose strict protection requirements for any data controllers including efforts to prevent unauthorised access. Those failing their security responsibilities are subject to hefty fines of up to £8.7 million or 2% of global turnover, potentially combined with further “corrective powers” (UK Government, 2018, Articles 33-83, Recitals 75-88). Outside organisational attacks, individuals are commonly targets for identity or credit theft (National Cyber Security Centre, 2025); victims may have money taken from online bank accounts or access to personal accounts stolen, such as social media.

Changes to regulations under the Data Use and Access Act (DUAA) 2025 has provided additional “legal bases” for the use of personal data for research purposes (ICO, 2025), while government initiatives such as the NHS Federated Data Platform (NHS England, 2024) are producing some of the largest connected systems sharing personal information on a national scale. Organisations will continue to harvest and circulate more data requiring more sophisticated tools to ensure it remains within the right hands.

What are the Benefits of the Solution?

Integrating machine learning techniques, making use of open-source models, provides a cost effective yet powerful solution to detect suspicious content. The inherent nature of AI allows for an evolving model, capable of being refined and retrained to continue combatting the latest threats. Public polling demonstrates a shared concern over online fraud, with 22% of people feeling at risk of “being tricked into giving money” (UK Home Office, 2025), further investigation confirms a distrusting zeitgeist toward institutional protection f

Does your project involve, or have any effect on human participants?

No

Does your proposed research involve any action on, or action that might affect any animals and/or plants?

No

Does your research involve the collection of data/information?

Yes

Please share how you plan to store and manage this data/information?

This project operates under the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018. All datasets used in this project are publicly available, open-source corpora sourced from their official repositories. Licencing terms of every dataset will be read in full and documented before use, ensuring use for academic research is permitted. Email communications contain real human data, with many datasets still including Personally Identifiable Information (PII). As documented in the project's data readiness process, automated PII scanning procedure will be applied to all text-based datasets prior to training. Any record in which personal identifiers are detected will be excluded entirely. All data will be stored on an encrypted local device or university-managed secure storage and deleted upon project completion in line with DPA 2018 regulations.

Data Readiness

First and foremost all candidate datasets must be suitable, legally permissible, and ethically acceptable for use in this project. A dataset that fails any assessment criterion will be substituted or escalated for supervisor review.

Suitability assessment evaluates the usefulness of the data, specifically looking for quality URL training features or a variety of social engineering strategies in text, or both. Another important factor is temporal relevance; to ensure experiment results accurately reflect modern phishing attacks, data must be recent.

Legal assessment confirms the dataset is permitted for academic research use, restrictions are understood and respected, and academic citation is recorded. Open source datasets may include licensing agreements. Terms can vary significantly across publicly available datasets and must be adhered to.

Ethical assessment ensures data originates from real human communications, as to avoid contaminating the experiment. Legal basis for its original collection must be justified and any PII for individuals cannot be present. Datasets involving real human communication are treated as sensitive by default (ICO, 2025), regardless of their public availability, and are subject to further PII assessment.

Storage assessment mandates the required storage can be accommodated on an encrypted local device or university-managed secure storage. Cloud storage services disrupt the chain of data custody over to third party providers and must not be used.

Following initial candidacy datasets can be acquired from its official source only. Third-party mirrors will not be used. At this point the data must be scrutinised to guarantee PII is not included in the data. Applying a checksum ensures provided file sizes and record counts are verified against the dataset documentation.

PII scanning is applied to any dataset containing natural language text. Automated scanning tools such as Microsoft Presidio (Microsoft, n.d.) can be used to identify PI including names, email addresses, phone numbers, financial account identifiers, and government-issued identification numbers. Any record containing PII is excluded from the dataset entirely.

For anomaly layer datasets where no natural language text is present, processes remain vigilant for PII; beneficial datasets may include hybrid data, if being used exclusively for URL feature extraction, any unrelated data excluded beforehand. The absence of raw personal data can be confirmed by inspecting the schema.

Could the storage, computations, or data analysis, whether through AI or standard coding, potentially harm individuals or organizations through data breaches, misuse, or unintended computational outcomes?

Yes

For projects that involve the use of AI, is it possible that either the outcomes, or the analysis, will have any impact on human decision making?

Yes

For projects that involve the use of AI, is it possible that the AI system could solely, that is without human intervention, make decisions pertaining to individuals?

Yes

Dr Pat Ryser-Welch

Associate lecturer - T192, T193, T194, M269, TM351, TM352, TM356, TM358, TM470

TM358 module team

Referred as: Her/She

Let's remember:

Start by doing what's necessary; then do what's possible; and suddenly you are doing the impossible.

Francis of Assisi

Mankind remains the most extraordinary computer of all....

John F. Kennedy

It does not matter how slowly you go as long as you do not stop.

Confucius

Read me : <https://interestingcomputing.com>